

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Shruti Sinha
Roll Number	MRA2026006
Programme / Department	Robotics and Artificial Intelligence
Topic ID	T13
Full Assigned Topic	Planning-Quality Degradation in Long-Horizon Autonomous Research Agents
AI Tool Used	
Date	

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations

B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☒ Occasionally ☐ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☒ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☒ Always ☐ Sometimes ☐ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☒ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	ChatGPT
Model name	
Model/version shown	
Access tier	☒ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☒ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☐ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☐ Yes ☐ No
Date/time generated	

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: “[YOUR ASSIGNED TOPIC]”. The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Planning-Quality Degradation in Long-Horizon Autonomous Research Agents

Measure	Value
Approximate pages	7
Approximate word count	2763
Number of sections	9
Total number of references	18
Approximate in-text citation occurrences	11
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☒ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	3
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	15
URL only	0
No persistent identifier supplied	0

TOTAL	
-------	--

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	6
R5	9
R6	12
R7	15
R8	16
R9	17
R10	18

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious

3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Looks Genuine	4	Yes
R2	Somewhat suspicious	2	Yes
R3	Somewhat suspicious	2	Yes
R4	Unsure	3	Yes
R5	Unsure	3	Yes
R6	Unsure	3	Yes
R7	Unsure	3	Yes
R8	Somewhat suspicious	2	Yes
R9	Somewhat suspicious	2	Yes
R10	Looks Genuine	4	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

☒ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	N	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	Y	Y	Y	Y
R8	Y	Y	Y	Y	Y	Y
R9	Y	Y	Y	Y	Y	Y
R10	Y	Y	Y	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	DOP	DOP	All information were correct
R2	arxiv	Arxiv	All information were correct
R3	Arxiv	Arxiv	All information were correct
R4	Arxiv	Arxiv	Year was not matching
R5	Arxiv	Arxiv	All information were correct
R6	Arxiv	Arxiv	All information were correct
R7	Arxiv	Arxiv	All information were correct
R8	Arxiv	Arxiv	All information were correct
R9	Arxiv	Arxiv	All information were correct
R10	arxiv	Arxiv	All information were correct

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	All information were correct
R2	A	All information were correct
R3	A	All information were correct
R4	B	Year of publication was given wrong
R5	A	All information were correct
R6	A	All information were correct
R7	A	All information were correct
R8	A	All information were correct

R9	A	All information were correct
R10	A	All information were correct

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 9 B = 1 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = $4A + 3B + C + E$

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

My calculation: $4(\underline{9}) + 3(\underline{1}) + 1(\underline{0}) + 0(\underline{0}) + 1(\underline{0}) = \underline{39}$ points

Authenticity Score = $39 / (4 \times 10) \times 100 = 97.5/100$

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = $4(6)+3(2)+1(1)=31$; maximum = 40; Authenticity Score = $31/40 \times 100 = 77.5$.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = $(\text{Correct genuine/fake predictions} / \text{Total predictions}) \times 100$

Correct predictions = 9 Total predictions = 10 Prediction Accuracy = 90 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	18
References deeply audited	10
A: Verified	9
B: Wrong metadata	1
C: Frankenstein	
D: Fabricated	

E: Identifier mismatch	
Authenticity Score	97.5 /100
Prediction Accuracy	90 %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

No. The audit showed that professional-looking citations are not automatically genuine. Although 9 out of the 10 deeply audited references were verified, 1 reference had incorrect metadata. This demonstrates that citations can appear credible while still containing errors.

2. Did all genuine references actually support the claims for which they were cited?

No, not necessarily. A reference can be genuine and still not adequately support the specific claim for which it is cited. Therefore, both the authenticity of the reference and its relevance to the claim must be checked

3. What was the most serious citation-integrity problem you observed?

The most serious problem was the presence of an incorrect reference among the citations. The audit identified **1 reference with wrong metadata**, showing that citation details should be independently verified rather than accepted at face value.

4. What is the single most important lesson you learned from this lab?

The most important lesson is that **AI-generated citations must always be independently verified**. A citation should be checked for its existence, bibliographic details, and whether it genuinely supports the claim being made. A high authenticity score is encouraging, but it does not eliminate the need for careful citation auditing

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: Shruti Sinha

Roll Number: MRA2026006



Signature:

Date: 21st August 2026

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support
Verification -> Quantitative Scores -> Submission



*Generate a scholarly research paper on the topic:
"*Planning-Quality Degradation in Long-Horizon Autonomous Research Agents ". *The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support*

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Planning-Quality Degradation in Long-Horizon Autono...

Planning-Quality Degradation in Long-Horizon Autonomous Research Agents

Abstract

Large language models (LLMs) have developed from passive text generators into increasingly autonomous agents capable of browsing the web, invoking software tools, maintaining memory, decomposing objectives, and executing multi-step workflows. These capabilities are particularly promising for autonomous research agents, whose tasks may involve discovering literature, formulating search strategies, collecting evidence, reconciling conflicting sources, synthesizing findings, and producing a defensible research report. However, performance that is strong on individual reasoning steps does not necessarily translate into reliable performance over long horizons. This paper examines **planning-quality degradation**: the systematic deterioration of an agent's ability to maintain coherent, evidence-grounded, goal-directed plans as the number of interdependent actions increases.

The paper synthesizes research on LLM planning, tool use, reasoning-and-acting, reflection, memory, long-context processing, autonomous-agent benchmarks, and web-based research. Existing evidence indicates several interacting degradation mechanisms: local or myopic decision-making, error propagation, context overload,

imperfect memory retrieval, goal drift, unreliable tool selection, premature commitment, weak verification, and vulnerability to environmental perturbations. Benchmarks such as WebArena, AgentBench, GAIA, BrowseComp, and τ -bench demonstrate that realistic multi-step tasks remain substantially more difficult than isolated question answering. Approaches including ReAct, Tree of Thoughts, LLM+P, Reflexion, skill libraries, external memory, and structured research workflows mitigate individual failure modes but do not yet provide a unified solution.

The paper argues that future autonomous research systems should treat planning as a continuously evaluated process rather than a one-time decomposition of a user objective. Promising directions include hierarchical and adaptive planning, explicit state representations, plan-quality estimators, uncertainty-aware search, evidence provenance graphs, execution checkpoints, selective replanning, and evaluation metrics that measure degradation as a function of horizon length. A central research challenge is therefore not merely increasing model intelligence, but developing agent architectures whose planning quality remains stable as tasks become longer, more uncertain, and more dynamically interactive.

Keywords: autonomous agents; large language models; long-horizon planning; planning quality; research agents; tool use; memory; reasoning; agent evaluation; information retrieval

1. Introduction

The emergence of large language model (LLM)-based autonomous agents has changed the scope of machine-assisted problem solving. Rather than producing an answer from a single prompt, an agent can now formulate intermediate objectives, search external information, invoke tools, inspect their outputs, revise its approach, and execute actions over an extended trajectory. Surveys of LLM-based autonomous agents describe this development as a transition from conventional language generation toward systems capable of perception, reasoning, planning, action, and feedback-driven adaptation (Wang et al., 2023; Huang et al., 2024).

Autonomous research is a particularly demanding application. A research task rarely consists of one information-retrieval operation. Instead, an agent may need to interpret a broad research question, identify subquestions, locate authoritative sources, distinguish primary from secondary evidence, compare competing findings, resolve contradictions, maintain citations, and synthesize the evidence into a coherent conclusion. The resulting workflow is both **long-horizon** and **state-**

dependent: later decisions depend on what was discovered earlier, while early mistakes can constrain the quality of all subsequent steps.

This creates an important distinction between *reasoning quality at a single step* and *planning quality across a trajectory*. An LLM may correctly reason about an individual document yet choose an ineffective search query. It may retrieve an appropriate source but fail to remember why that source was relevant several steps later. It may generate a plausible intermediate conclusion and then unconsciously use that conclusion to bias subsequent searches. Thus, an agent can exhibit high local competence while producing a globally poor research process.

This phenomenon is referred to in this paper as **planning-quality degradation (PQD)**. PQD is defined as the decline in the quality, coherence, relevance, efficiency, or reliability of an agent's planned behavior as the temporal horizon and dependency structure of a task increase. The concept is broader than ordinary execution failure. An execution failure occurs when an action does not achieve its intended effect; planning degradation occurs when the agent's representation of *what should be done next and why* becomes progressively less aligned with the overall objective.

Several findings motivate this perspective. AgentBench identified poor long-term reasoning, decision-making, and instruction following as important obstacles for usable LLM agents across multiple interactive environments (Liu et al., 2024). WebArena demonstrated that realistic web tasks involving multiple interactions remain difficult even for powerful models; its reported GPT-4-based baseline achieved only 14.41% end-to-end success compared with 78.24% human performance (Zhou et al., 2024). Similarly, GAIA showed a large gap between human performance and GPT-4 with plugins on real-world assistant tasks requiring reasoning, browsing, multimodality, and tool use (Mialon et al., 2024).

The central argument of this paper is that long-horizon autonomous research should be understood as a **planning stability problem** as much as a reasoning problem. Increasing model scale or context length can improve individual capabilities, but these improvements do not automatically guarantee stable planning over hundreds of dependent actions. Indeed, research on long-context language modeling has shown that simply providing a model with more context does not ensure that it will reliably use all relevant information (Liu et al., 2024).

The remainder of the paper reviews the foundations of LLM-agent planning, characterizes the principal mechanisms underlying planning-quality degradation, examines current mitigation strategies, and identifies research directions for building

autonomous research agents that remain coherent and evidence-grounded over extended horizons.

2. Background and Related Work

2.1 LLM Agents as Planning Systems

Traditional autonomous-agent research conceptualizes an agent as a system that observes an environment, maintains a state representation, selects actions, and seeks to maximize an objective. LLM agents retain this basic structure but replace or augment conventional symbolic planners, policies, or controllers with a language model capable of interpreting natural-language objectives and generating actions.

Wang et al. (2023) provide a broad survey of LLM-based autonomous agents and identify recurring components including profiling, memory, planning, action generation, and interaction with environments. Huang et al. (2024) specifically organize LLM-agent planning research around task decomposition, plan selection, external modules, reflection, and memory. This taxonomy is useful because planning quality depends not only on the model's intrinsic reasoning ability but also on how these components interact.

A simplified formulation of an agent's planning problem is:

$$[P_t = \pi(G, S_t, M_t, E_t, \Theta),]$$

where (G) represents the global research goal, (S_t) the current environment state, (M_t) the agent's memory, (E_t) the available evidence and tools, and (Θ) the model and prompting configuration. The resulting plan (P_t) determines one or more future actions.

In a long-horizon setting, however, (S_t), (M_t), and (E_t) evolve after every action. Therefore,

$$[P_{t+1} \neq P_t]$$

in general. A robust agent should update its plan when new evidence changes the expected value of future actions. Planning quality consequently depends on **continual alignment between the global objective and local decisions.**

2.2 Reasoning, Acting, and Tool Use

ReAct introduced an influential architecture in which reasoning traces and external actions are interleaved. Rather than producing a complete plan before execution, the model reasons about the current state, performs an action, observes the result, and reasons again (Yao et al., 2023). The approach improved performance on interactive tasks because environmental observations can correct assumptions made during reasoning.

Tool use further extends the agent's effective capabilities. Toolformer demonstrated that language models can learn when to call external APIs and how to incorporate returned information into subsequent generation (Schick et al., 2023). WebGPT similarly showed how browser interaction and citation collection can improve web-based question answering (Nakano et al., 2022).

For research agents, however, tool use introduces another source of planning complexity. The agent must not only decide *what information is needed*, but also *which tool should obtain it, when it should be used, how its output should be interpreted, and whether the resulting evidence is trustworthy*. Consequently, tool invocation becomes part of the planning problem rather than a separate implementation detail.

2.3 Explicit Search and Deliberation

Tree of Thoughts (ToT) extended chain-of-thought reasoning by allowing models to explore multiple intermediate reasoning paths, evaluate alternatives, and backtrack when necessary (Yao et al., 2023). On the Game of 24 task, the authors reported a substantial improvement over conventional chain-of-thought prompting, illustrating the value of explicit search for problems where early decisions strongly affect later outcomes.

This principle is particularly relevant to research. A research agent may face several possible interpretations of a question or several plausible search strategies. Committing immediately to one strategy can lead to a narrow evidence base. Deliberative planning can instead preserve multiple hypotheses until sufficient evidence is obtained.

2.4 External Planning and Classical Planners

LLM+P provides an alternative approach: use the LLM for natural-language understanding and translation while delegating formal planning to a classical planner. Liu et al. (2023) convert natural-language descriptions into Planning Domain Definition Language (PDDL), invoke a classical planner, and translate the resulting

plan back into natural language. Their experiments showed that classical planners could generate feasible or optimal plans for tasks where LLMs frequently failed to produce feasible long-horizon plans.

This result highlights an important architectural principle: **language competence and planning competence are not identical capabilities**. An autonomous research system may benefit from delegating portions of planning to specialized modules rather than expecting one generative model to perform every function.

2.5 Reflection and Memory

Reflexion introduced verbal reinforcement learning in which an agent converts feedback from previous attempts into linguistic reflections stored in an episodic memory buffer (Shinn et al., 2023). The approach improved performance across sequential decision-making, coding, and reasoning tasks without modifying model weights.

Memory is particularly important for long-horizon research because an agent may accumulate hundreds of observations, sources, intermediate conclusions, and failed searches. Yet memory itself can become a source of degradation. Irrelevant or incorrect memories may be retrieved later, while important information may be buried in a large context. Thus, the problem is not simply *whether an agent has memory*, but whether its memory provides an accurate and task-relevant state representation.

3. Planning-Quality Degradation in Long-Horizon Research

3.1 Defining Planning Quality

Planning quality can be considered multidimensional. For a research trajectory ($T = (a_1, \dots, a_H)$) of horizon (H), let:

$$Q(T) = \alpha C + \beta E + \gamma R + \delta V - \lambda K,$$

where:

- (C) = coherence with the global research objective;
- (E) = evidence quality and coverage;
- (R) = relevance of actions and retrieved information;
- (V) = verification and factual reliability;

- (K) = unnecessary cost, including redundant searches and tool calls.

As (H) increases, these quantities may change. A planning-degradation curve can therefore be represented conceptually as:

$$[\\ D(H)=Q(H_0)-Q(H), \\]$$

where (H₀) is a short-horizon baseline. A reliable long-horizon agent should minimize (D(H)) as the horizon increases.

This formulation emphasizes that long-horizon failure is not necessarily binary. An agent may eventually complete a task while becoming increasingly inefficient, relying on weaker evidence, repeating searches, or abandoning parts of the original objective. Such degradation can remain hidden when evaluation reports only final task success.

3.2 Myopic Decision-Making

The first major mechanism is **myopia**. Step-by-step reasoning often evaluates the immediate usefulness of an action rather than its long-term consequences. For example, a research agent may select a search query because it produces many superficially relevant results, even though another query would provide more authoritative primary literature.

This is closely related to the distinction between local and global optimization. If an agent selects

$$[\\ a_t=\arg\max_a U(s_t,a), \\]$$

it may choose the action with the greatest immediate utility. A long-horizon planner instead seeks something closer to:

$$[\\ a_t^*=\arg\max_a \mathbb{E}\left[\sum_{k=t}^H \gamma^{k-t}U(s_k,a_k)\right]. \\]$$

The latter requires reasoning about delayed consequences.

Recent planning-centric analysis argues explicitly that step-wise reasoning can behave like a greedy policy and that this becomes problematic as the planning

horizon increases (Wang et al., 2026). This emerging literature strengthens the case for treating planning as a distinct capability rather than assuming that better reasoning automatically produces better long-horizon plans.

3.3 Error Propagation and Compounding

Long-horizon agents are exposed to **compounding errors**. Suppose each dependent step has probability (p) of being correct. Under a simplified independence assumption, the probability that all (H) steps are correct is:

$$P(\text{success}) = p^H.$$

Even with ($p=0.98$), a 100-step trajectory has an idealized probability of approximately $(0.98^{100}) \approx 0.133$ of being error-free. Real systems are more complicated because errors are neither independent nor equally consequential, but the equation illustrates why small per-step weaknesses can become severe over long trajectories.

Research tasks are especially vulnerable because one early mistaken assumption may influence search queries, source selection, intermediate summaries, and final synthesis. ReAct partially addresses this problem by incorporating environmental observations into the reasoning loop, while Reflexion attempts to learn from failures across episodes (Yao et al., 2023; Shinn et al., 2023).

3.4 Context Overload and State Loss

Long-horizon research produces a rapidly expanding context containing search results, source excerpts, notes, intermediate hypotheses, tool outputs, and prior plans. More context is not equivalent to more usable information.

Liu et al. (2024) demonstrated the "lost in the middle" phenomenon, finding that language-model performance can decrease when relevant information appears in the middle of long contexts, with stronger performance when relevant material occurs near the beginning or end.

For research agents, this can manifest as **state loss**. The agent technically retains earlier evidence but fails to retrieve or appropriately weight it. Consequently, it may rediscover information, contradict earlier findings, or make decisions based disproportionately on recent observations.

The distinction between *long context* and *effective memory* is therefore fundamental. Increasing the context window may reduce truncation but does not solve retrieval, prioritization, or state abstraction.

3.5 Goal Drift

Goal drift occurs when an agent's behavior gradually moves away from the original objective. It may begin with “compare the evidence for competing hypotheses” and eventually behave as though the objective were simply “collect information supporting hypothesis A.”

Goal drift can arise through several mechanisms:

1. **Query-induced narrowing:** early search results disproportionately influence subsequent searches.
2. **Confirmation bias:** intermediate conclusions become implicit assumptions.
3. **Local completion pressure:** the agent prematurely interprets a partially answered subproblem as the overall solution.
4. **Memory distortion:** summaries omit qualifications and preserve conclusions without context.
5. **Tool-induced framing:** available tools constrain what the agent considers possible.

Because research is exploratory, these effects are especially important. An autonomous research system must preserve uncertainty rather than prematurely collapsing a hypothesis space.

3.6 Tool-Use Degradation

Tool use expands the action space and therefore increases planning complexity. An agent may choose among search engines, browsers, databases, code execution, document parsers, calculators, APIs, and specialized retrieval systems.

The problem is not merely selecting a tool but selecting the *appropriate sequence of tools*. Tool errors can also alter the state of the research process. AgentDojo demonstrates that agents operating over external tool outputs can be vulnerable to prompt injection, while even benign tasks remain challenging for state-of-the-art systems (Debenedetti et al., 2024).

For a research agent, untrusted web content can therefore affect not only factual correctness but also the future plan. A malicious or misleading document may cause the agent to change its objective, reveal sensitive information, or waste subsequent actions. Tool reliability and planning robustness are consequently coupled.

3.7 Premature Commitment

Research often requires maintaining multiple possible explanations until sufficient evidence is collected. An agent that commits too early can generate a coherent but poorly supported narrative.

Tree of Thoughts provides one mechanism for maintaining multiple candidate reasoning trajectories (Yao et al., 2023). Classical planning approaches likewise demonstrate the value of search over candidate plans rather than relying solely on sequential generation (Liu et al., 2023).

For research agents, premature commitment should be measured explicitly. A system should be evaluated on whether it can revise a hypothesis after encountering contradictory evidence, not merely whether it can produce a plausible final answer.

4. Current Approaches and Developments

4.1 Receding-Horizon Planning

A practical strategy is to avoid generating an extremely long plan in advance. Instead, the agent generates a short sequence, executes it, observes the environment, and replans. This resembles receding-horizon control.

ReAct follows a related principle by interleaving reasoning and action. For research agents, receding-horizon planning can prevent stale plans from surviving after new evidence changes the problem.

Its limitation is that overly short planning windows may themselves produce myopic behavior. The challenge is therefore to determine an appropriate planning depth dynamically.

4.2 Hierarchical Planning

Hierarchical planning decomposes a broad objective into increasingly specific subgoals:

[
 $G \rightarrow \{G_1, G_2, \dots, G_n\}$
]

and then:

[
 $G_i \rightarrow \{g_{i1}, g_{i2}, \dots\}$.
]

For autonomous research, the top level might define research questions, the middle level define evidence categories, and the lower level define individual retrieval and verification operations.

Hierarchical planning limits the cognitive burden of simultaneously representing an entire task while preserving global structure. However, decomposition can also introduce errors if the chosen subgoals are incomplete or mutually inconsistent.

4.3 Search-Based Deliberation

Tree of Thoughts demonstrates that search over candidate reasoning paths can improve performance when decisions require lookahead and backtracking. The broader implication is that autonomous agents should not always behave as single-path generators.

For research systems, candidate plans could be evaluated according to source diversity, expected information gain, authority of evidence, cost, and probability of resolving the research question.

4.4 External Planners

LLM+P illustrates a hybrid design in which the LLM translates natural-language objectives into a representation suitable for classical planning, allowing established search algorithms to determine a plan.

Such hybridization is attractive because different components can specialize:

- LLMs: semantic interpretation and flexible language interaction;
- retrieval systems: information access;
- classical planners: constraint satisfaction and search;
- databases: persistent structured state;
- verification modules: factual consistency;
- execution controllers: reliable tool invocation.

The major challenge is constructing interfaces that preserve semantic information across these modules.

4.5 Reflection and Self-Correction

Reflexion demonstrates that agents can store textual feedback from unsuccessful trajectories and use it during subsequent attempts. Reflection is valuable because long-horizon tasks inevitably encounter errors.

However, self-reflection is not equivalent to reliable correction. A model may generate a plausible explanation for failure without identifying the actual causal mistake. Future systems therefore require externally grounded verification in addition to linguistic self-critique.

4.6 Persistent Skill and Memory Libraries

Voyager uses an automatic curriculum, an executable skill library, and iterative prompting based on environmental feedback. The system demonstrates how previously acquired skills can support future behavior in an open-ended environment (Wang et al., 2024).

For research agents, analogous memory structures could store:

- successful search strategies;
- source-quality patterns;
- domain-specific terminology;
- validated analytical procedures;
- reusable evidence schemas;
- failed approaches and their causes.

The key research question is how to prevent accumulated memories from becoming contradictory, obsolete, or misleading.

4.7 Improved Evaluation Environments

Modern benchmarks increasingly expose long-horizon weaknesses that are invisible in single-turn question answering.

WebArena evaluates realistic web tasks involving multiple websites and interactions; its relatively low agent success rate illustrates the difficulty of realistic web navigation. AgentBench evaluates agents across eight interactive environments and identifies long-term reasoning and decision-making as important obstacles. GAIA focuses on general assistant tasks requiring reasoning, multimodality, web browsing, and tool use.

More recent benchmarks increasingly emphasize persistence and reliability. BrowseComp contains 1,266 difficult browsing problems requiring persistent navigation and creative information discovery (Wei et al., 2025). τ -bench evaluates

dynamic tool-agent-user interactions and reports substantial inconsistency across repeated trials, illustrating that average success alone is insufficient for evaluating reliable autonomy (Yao et al., 2025).

5. Research Challenges and Limitations

5.1 Measuring Degradation Rather Than Final Success

Most agent benchmarks report whether a task was completed. This obscures *when* and *why* planning quality deteriorated.

A better evaluation protocol should measure:

[
Q₁, Q₂, \ldots, Q_H
]

throughout the trajectory. Useful metrics could include:

- goal alignment at each step;
- evidence coverage;
- source authority;
- redundancy;
- number of unnecessary actions;
- plan revisions;
- contradiction rate;
- memory-retrieval accuracy;
- tool-selection accuracy;
- recovery time after errors.

The resulting **planning degradation curve** would provide more diagnostic information than end-to-end success alone.

5.2 Attribution of Failure

An unsuccessful research task can fail because of retrieval, reasoning, planning, memory, tool execution, or verification. These causes are often entangled.

For example, if an agent selects a poor source, was the problem:

1. failure to understand the question;
2. poor search planning;
3. inadequate retrieval;

4. incorrect source evaluation;
5. memory loss; or
6. confirmation bias?

Benchmarks therefore need causal diagnostic annotations rather than merely final labels.

5.3 Uncertainty Calibration

Long-horizon research requires the agent to know when it lacks sufficient evidence. Current language models can produce confident outputs even when the evidence is incomplete.

An autonomous researcher should distinguish among:

[
\text{known}, \quad \text{supported}, \quad \text{uncertain}, \quad \text{contradictory}.
]

This distinction should be preserved throughout planning rather than added only during final answer generation.

5.4 Dynamic and Adversarial Environments

Research environments are non-stationary. Web pages change, search rankings evolve, documents disappear, APIs fail, and retrieved content may contain misleading instructions. AgentDojo illustrates the security consequences of tool-mediated environments containing untrusted information.

A robust planner must therefore optimize not only for task completion but also for resilience to environmental change.

5.5 Cost and Computational Scaling

Long-horizon deliberation can be computationally expensive. Tree search, repeated reflection, multiple model calls, and verification all increase latency and cost.

The ideal architecture should therefore allocate computation selectively. Simple decisions should be handled cheaply, while ambiguous or high-impact decisions should receive deeper search and verification.

5.6 Benchmark Contamination and Reproducibility

Autonomous research benchmarks are especially vulnerable to data leakage because web-based information can enter model training corpora. BrowseComp explicitly addresses this concern by designing questions around difficult-to-find information and using measures intended to reduce direct memorization.

Future evaluations should use dynamic or procedurally generated tasks, held-out sources, and temporal splits to distinguish genuine research ability from memorization.

6. Research Gaps and Future Directions

6.1 A Formal Theory of Planning Degradation

A major gap is the absence of a widely accepted mathematical theory of planning-quality degradation in LLM agents. Existing work identifies individual problems—memory loss, myopia, tool failure, long-context degradation, or reflection errors—but these mechanisms are rarely unified.

Future research should model planning quality as a function of horizon, uncertainty, branching factor, memory load, and environmental volatility:

$$[\\ Q = f(H,U,B,M,V). \\]$$

Such a framework could enable researchers to predict when a particular agent architecture is likely to become unreliable.

6.2 Horizon-Adaptive Planning

Agents should dynamically select planning depth. When the next action is obvious, shallow reasoning is sufficient. When an early choice has large downstream consequences, the system should invoke deeper lookahead.

A possible policy is:

$$[\\ d_t = f(\text{uncertainty}, \text{impact}, \text{branching}, \text{horizon}), \\]$$

where (d_t) represents the amount of planning computation allocated at step (t).

This would combine the efficiency of reactive agents with the robustness of deliberate planners.

6.3 Evidence-Centric Research Planning

Autonomous research agents should represent evidence explicitly rather than treating retrieved text as an undifferentiated context.

A promising representation is an **evidence graph**:

```
[  
G_E=(V,E),  
]
```

where nodes (V) represent claims, sources, datasets, methods, and observations, while edges (E) represent relations such as *supports*, *contradicts*, *derived from*, and *requires verification*.

Planning could then be guided by unresolved nodes—for example, identifying which claim has insufficient evidence or which contradiction requires further investigation.

6.4 Provenance-Aware Memory

Memory systems should preserve not only information but provenance. Instead of storing:

“Study X shows method A performs better.”

the agent should store:

claim → source → evidence location → context → confidence → date.

This would reduce the risk of unsupported summaries becoming persistent false beliefs.

6.5 Plan Criticism by Independent Evaluators

Self-criticism is limited when the same model generates both the plan and the critique. Future systems could use independent planning critics, specialized verification models, symbolic constraints, or executable tests.

For example:

```
[
\text{Planner} \rightarrow \text{Plan} \rightarrow
\{\text{Verifier}_1, \text{Verifier}_2, \text{EvidenceChecker}\}
\rightarrow \text{Replanner}.
]
```

The planner would only proceed when critical constraints are satisfied.

6.6 Research-Specific Long-Horizon Benchmarks

Existing benchmarks measure important components but do not fully capture scientific research. Future benchmarks should require agents to:

1. formulate a research plan;
2. search across heterogeneous scholarly sources;
3. distinguish primary and secondary evidence;
4. reconcile conflicting findings;
5. track citations and provenance;
6. revise hypotheses after contradictory evidence;
7. produce a final synthesis;
8. justify why the selected evidence is sufficient.

Evaluation should score the entire trajectory, not only the final prose.

6.7 Longitudinal Evaluation

A particularly important gap is repeated evaluation. τ -bench demonstrates that agents can exhibit substantial variability across repeated trials. Long-horizon research systems should therefore be evaluated over many independent runs.

A reliable system should exhibit not only high mean performance but also low variance:

```
[
\text{Reliability} \approx
\mathbb{E}[Q] - \lambda \operatorname{Var}(Q).
]
```

This shifts evaluation from “Can the agent solve the task once?” to “Can the agent reliably solve comparable tasks without unpredictable planning failures?”

6.8 Human-in-the-Loop Checkpoints

Fully autonomous research may not be optimal for every horizon. Strategic human checkpoints could be introduced at high-impact decisions such as research-question interpretation, hypothesis selection, evidence sufficiency, and final synthesis.

The goal is not to replace autonomy but to allocate human oversight where the expected cost of planning failure is greatest.

7. Conclusion

Long-horizon autonomous research exposes a fundamental limitation of current LLM-agent systems: strong local reasoning does not guarantee stable global planning. As an agent performs more actions, it must continually preserve its objectives, retrieve relevant memory, evaluate evidence, select appropriate tools, detect errors, revise hypotheses, and decide whether its current plan remains valid. Failure in any of these processes can compound over time.

The literature already provides strong evidence for individual components of this problem. ReAct demonstrates the value of grounding reasoning in environmental feedback; Tree of Thoughts shows the benefits of explicit search and backtracking; LLM+P demonstrates the value of delegating planning to formal planners; Reflexion illustrates how feedback and memory can improve future behavior; and Voyager demonstrates the potential of persistent skill libraries. Meanwhile, WebArena, AgentBench, GAIA, BrowseComp, and τ -bench reveal that realistic, multi-step autonomy remains substantially harder than isolated language reasoning.

The central research challenge is therefore not simply to make models “reason better.” It is to construct systems whose **planning quality remains stable as the horizon grows**. This requires explicit state representations, adaptive planning depth, evidence-aware memory, reliable verification, uncertainty calibration, selective replanning, and evaluation methods that expose degradation throughout a trajectory.

For autonomous research agents, the ultimate standard should be more demanding than producing a plausible final report. A trustworthy system should be able to demonstrate that its research trajectory remained aligned with the original question, that important evidence was neither lost nor selectively ignored, that contradictory findings triggered appropriate replanning, and that conclusions can be traced to verifiable sources. Achieving this form of planning stability would represent an important step from capable language models toward genuinely reliable autonomous research systems.

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Planning-Quality Degradation in Long-Horizon Autonomous Research Agents

Abstract

Large language models (LLMs) have developed from passive text generators into increasingly autonomous agents capable of browsing the web, invoking software tools, maintaining memory, decomposing objectives, and executing multi-step workflows. These capabilities are particularly promising for autonomous research agents, whose tasks may involve discovering literature, formulating search strategies, collecting evidence, reconciling conflicting sources, synthesizing findings, and producing a defensible research report. However, performance that is strong on individual reasoning steps does not necessarily translate into reliable performance over long horizons. This paper examines planning-quality degradation: the systematic deterioration of an agent's ability to maintain coherent, evidence-grounded, goal-directed plans as the number of interdependent actions increases.

The paper synthesizes research on LLM planning, tool use, reasoning-and-acting, reflection, memory, long-context processing, autonomous-agent benchmarks, and web-based research. Existing evidence indicates several interacting degradation mechanisms: local or myopic decision-making, error propagation, context overload, imperfect memory retrieval, goal drift, unreliable tool selection, premature commitment, weak verification, and vulnerability to environmental perturbations. Benchmarks such as WebArena, AgentBench, GAIA, BrowseComp, and τ -bench demonstrate that realistic multi-step tasks remain substantially more difficult than isolated question answering. Approaches including ReAct, Tree of Thoughts, LLM+P, Reflexion, skill libraries, external memory, and structured research workflows mitigate individual failure modes but do not yet provide a unified solution.

The paper argues that future autonomous research systems should treat planning as a continuously evaluated process rather than a one-time decomposition of a user objective. Promising directions include hierarchical and adaptive planning, explicit state representations, plan-quality estimators, uncertainty-aware search, evidence provenance graphs, execution checkpoints, selective replanning, and evaluation metrics that measure degradation as a function of horizon length. A central research challenge is therefore not merely increasing model intelligence, but developing agent architectures whose planning quality remains stable as tasks become longer, more uncertain, and more dynamically interactive.

Keywords: autonomous agents; large language models; long-horizon planning; planning quality; research agents; tool use; memory; reasoning; agent evaluation; information retrieval

1. Introduction

The emergence of large language model (LLM)-based autonomous agents has changed the scope of machine-assisted problem solving. Rather than producing an answer from a single prompt, an agent can now formulate intermediate objectives, search external information, invoke tools, inspect their outputs, revise its approach, and execute actions over an extended trajectory. Surveys of LLM-based autonomous agents describe this development as a transition from conventional language generation toward systems capable of perception, reasoning, planning, action, and feedback-driven adaptation (Wang et al., 2023; Huang et al., 2024).

Autonomous research is a particularly demanding application. A research task rarely consists of one information-retrieval operation. Instead, an agent may need to interpret a broad research question, identify subquestions, locate authoritative sources, distinguish primary from secondary evidence, compare competing

findings, resolve contradictions, maintain citations, and synthesize the evidence into a coherent conclusion. The resulting workflow is both long-horizon and state-dependent: later decisions depend on what was discovered earlier, while early mistakes can constrain the quality of all subsequent steps.

This creates an important distinction between reasoning quality at a single step and planning quality across a trajectory. An LLM may correctly reason about an individual document yet choose an ineffective search query. It may retrieve an appropriate source but fail to remember why that source was relevant several steps later. It may generate a plausible intermediate conclusion and then unconsciously use that conclusion to bias subsequent searches. Thus, an agent can exhibit high local competence while producing a globally poor research process.

This phenomenon is referred to in this paper as planning-quality degradation (PQD). PQD is defined as the decline in the quality, coherence, relevance, efficiency, or reliability of an agent's planned behavior as the temporal horizon and dependency structure of a task increase. The concept is broader than ordinary execution failure. An execution failure occurs when an action does not achieve its intended effect; planning degradation occurs when the agent's representation of what should be done next and why becomes progressively less aligned with the overall objective.

Several findings motivate this perspective. AgentBench identified poor long-term reasoning, decision-making, and instruction following as important obstacles for usable LLM agents across multiple interactive environments (Liu et al., 2024). WebArena demonstrated that realistic web tasks involving multiple interactions remain difficult even for powerful models; its reported GPT-4-based baseline achieved only 14.41% end-to-end success compared with 78.24% human performance (Zhou et al., 2024). Similarly, GAIA showed a large gap between human performance and GPT-4 with plugins on real-world assistant tasks requiring reasoning, browsing, multimodality, and tool use (Mialon et al., 2024).

The central argument of this paper is that long-horizon autonomous research should be understood as a planning stability problem as much as a reasoning problem. Increasing model scale or context length can improve individual capabilities, but these improvements do not automatically guarantee stable planning over hundreds of dependent actions. Indeed, research on long-context language modeling has shown that simply providing a model with more context does not ensure that it will reliably use all relevant information (Liu et al., 2024).

2. Background and Related Work

Traditional autonomous-agent research conceptualizes an agent as a system that observes an environment, maintains a state representation, selects actions, and seeks to maximize an objective. LLM agents retain this basic structure but replace or augment conventional symbolic planners, policies, or controllers with a language model capable of interpreting natural-language objectives and generating actions.

Wang et al. (2023) provide a broad survey of LLM-based autonomous agents and identify recurring components including profiling, memory, planning, action generation, and interaction with environments. Huang et al. (2024) specifically organize LLM-agent planning research around task decomposition, plan selection, external modules, reflection, and memory. ReAct interleaves reasoning and external actions, while Toolformer and WebGPT demonstrate the value of tool use and browser interaction (Yao et al., 2023; Schick et al., 2023; Nakano et al., 2022).

Tree of Thoughts (Yao et al., 2023) extends sequential reasoning by allowing multiple candidate reasoning paths, evaluation, and backtracking. LLM+P (Liu et al., 2023) instead delegates formal planning to classical planners after translating natural-language objectives into PDDL. These approaches establish that language competence and planning competence are related but distinguishable capabilities.

Reflexion (Shinn et al., 2023) introduces verbal reinforcement learning in which feedback from unsuccessful attempts is stored in an episodic memory buffer. Such memory is particularly important for long-horizon research, but memory can itself become a source of degradation when irrelevant or incorrect information is retrieved or when important evidence is buried in a large context.

3. Planning-Quality Degradation in Long-Horizon Research

Planning quality is multidimensional. For a research trajectory $T=(a_1, \dots, a_H)$ of horizon H , it can be conceptualized as $Q(T)=\alpha C+\beta E+\gamma R+\delta V-\lambda K$, where C represents coherence with the global research objective, E evidence quality and coverage, R relevance, V verification and factual reliability, and K unnecessary cost.

Several interacting mechanisms cause degradation. First, myopic decision-making can favor actions with high immediate utility over actions that improve the overall research trajectory. Recent planning-centric work argues that step-wise reasoning can behave like a greedy policy as the horizon increases (Wang et al., 2026).

Second, errors compound. Under a simplified independence assumption, if every dependent step has correctness probability p , then $P(\text{success})=p^H$. Even small per-step weaknesses can therefore become severe over long trajectories.

Third, context overload can produce effective state loss. Liu et al. (2024) documented the “lost in the middle” phenomenon, showing that models may fail to use relevant information reliably when it is embedded in long contexts. For research agents, this can result in repeated searches, contradictory summaries, or excessive dependence on recent information.

Fourth, goal drift can occur when early search results, intermediate conclusions, memory summaries, or tool affordances progressively narrow the agent's interpretation of the original research objective. Fifth, tool use enlarges the action space and introduces new failure modes. AgentDojo demonstrates that tool-mediated environments can expose agents to prompt injection and other security problems (Debenedetti et al., 2024).

Finally, premature commitment can cause a research agent to select a hypothesis or evidence path before sufficient evidence has been collected. Search-based methods such as Tree of Thoughts mitigate this by preserving multiple candidate trajectories.

4. Current Approaches and Developments

Current mitigation strategies include receding-horizon planning, hierarchical planning, search-based deliberation, external planners, reflection, persistent memory, and improved evaluation environments.

Receding-horizon approaches avoid generating extremely long plans in advance. The agent executes a short sequence, observes the environment, and replans. ReAct follows a related principle by interleaving reasoning and action. Hierarchical planning decomposes a global objective into research questions, evidence categories, and concrete retrieval or verification operations.

Search-based deliberation allows multiple candidate plans to be explored and evaluated. LLM+P demonstrates that classical planners can provide strong formal planning capabilities when connected to language models. Reflexion provides a mechanism for learning from previous failures, while Voyager demonstrates the potential of persistent executable skill libraries.

Evaluation has also advanced. WebArena evaluates realistic multi-site web tasks; AgentBench evaluates agents across multiple interactive environments; GAIA focuses on general assistant tasks requiring reasoning, multimodality, browsing, and tools. BrowseComp emphasizes persistent browsing and difficult

information discovery, while τ -bench evaluates dynamic tool-agent-user interaction and exposes variability across repeated trials.

5. Research Challenges and Limitations

A major limitation of current evaluation is its focus on final task success rather than the trajectory. A stronger evaluation should measure goal alignment, evidence coverage, source authority, redundancy, plan revisions, contradiction rate, memory retrieval, tool selection, and recovery time throughout the task.

Failure attribution is another challenge because retrieval, reasoning, planning, memory, tool execution, and verification errors are often entangled. Long-horizon research also requires uncertainty calibration: an agent should explicitly distinguish known, supported, uncertain, and contradictory claims.

Research environments are dynamic and potentially adversarial. Web pages change, APIs fail, documents disappear, and retrieved content can contain misleading instructions. Long-horizon deliberation is also computationally expensive because search, reflection, verification, and repeated model calls increase latency and cost.

Finally, benchmark contamination remains a concern because web-based benchmark information can enter training corpora. Dynamic tasks, held-out sources, temporal splits, and procedurally generated research questions can help distinguish genuine research ability from memorization.

6. Research Gaps and Future Directions

The field lacks a unified theory of planning-quality degradation in LLM agents. Future work should model planning quality as a function of horizon, uncertainty, branching factor, memory load, and environmental volatility.

Horizon-adaptive planning is promising: agents could allocate deeper planning to decisions with high uncertainty or downstream impact while handling routine decisions cheaply. Evidence-centric planning could represent claims, sources, datasets, methods, and observations in an evidence graph whose edges encode support, contradiction, derivation, and verification requirements.

Memory should become provenance-aware, preserving the source, evidence location, context, confidence, and date associated with each stored claim. Independent plan critics and verification modules could reduce the weakness of self-criticism by separating planning from evaluation.

Research-specific benchmarks should require agents to formulate a plan, search heterogeneous scholarly sources, distinguish primary and secondary evidence, reconcile contradictions, maintain citation provenance, revise hypotheses, and justify evidence sufficiency. Evaluation should score the entire trajectory and include repeated trials so that both mean performance and variance are measured.

Finally, human-in-the-loop checkpoints can be introduced at high-impact decisions such as research-question interpretation, hypothesis selection, evidence sufficiency, and final synthesis. The objective is not to eliminate autonomy but to allocate oversight where planning failure would be most costly.

7. Conclusion

Long-horizon autonomous research exposes a fundamental limitation of current LLM-agent systems: strong local reasoning does not guarantee stable global planning. As an agent performs more actions, it must continually preserve its objectives, retrieve relevant memory, evaluate evidence, select appropriate tools, detect errors, revise hypotheses, and decide whether its current plan remains valid.

The literature provides strong evidence for individual components of this problem. ReAct demonstrates the value of grounding reasoning in environmental feedback; Tree of Thoughts shows the benefits of explicit search and backtracking; LLM+P demonstrates the value of delegating planning to formal planners; Reflexion illustrates how feedback and memory can improve future behavior; and Voyager demonstrates the potential of persistent skill libraries.

The central research challenge is therefore not simply to make models reason better. It is to construct systems whose planning quality remains stable as the horizon grows. This requires explicit state representations, adaptive planning depth, evidence-aware memory, reliable verification, uncertainty calibration, selective replanning, and evaluation methods that expose degradation throughout a trajectory.

For autonomous research agents, the ultimate standard should be more demanding than producing a plausible final report. A trustworthy system should demonstrate that its research trajectory remained aligned with the original question, that important evidence was neither lost nor selectively ignored, that contradictory findings triggered appropriate replanning, and that conclusions can be traced to verifiable sources.

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